# ✅ Easy (10 Questions)

1.

**What is a virus in malware terminology?**
A) A self-replicating program that spreads by infecting files
B) Malware that encrypts files for ransom
C) Software that monitors keystrokes
D) Advertising software
**Answer:** A
**Explanation:** Viruses attach to files and spread when files are shared.

2.

**Which malware replicates itself without needing a host file?**
A) Worm
B) Trojan
C) Ransomware
D) Spyware
**Answer:** A
**Explanation:** Worms spread across networks independently.

3.

**What type of malware disguises as legitimate software?**
A) Trojan Horse
B) Worm
C) Virus
D) Keylogger
**Answer:** A
**Explanation:** Trojans trick users into installing them.

4.

**Ransomware is designed to:**
A) Encrypt user data and demand payment
B) Monitor user activity secretly
C) Spread through email attachments only
D) Display unwanted advertisements
**Answer:** A
**Explanation:** Ransomware locks data until ransom is paid.

5.

**Which malware type records keystrokes?**
A) Keylogger
B) Worm
C) Virus
D) Rootkit
**Answer:** A
**Explanation:** Keyloggers capture user input.

6.

**What is spyware mainly used for?**
A) Monitoring user activities and sending info to attackers
B) Encrypting files
C) Infecting other files
D) Displaying pop-up ads
**Answer:** A
**Explanation:** Spyware spies on users.

7.

**A rootkit is designed to:**
A) Hide malware presence by modifying system files
B) Encrypt files for ransom
C) Replicate through networks
D) Steal passwords by keylogging
**Answer:** A
**Explanation:** Rootkits conceal malicious software.

8.

**Fileless malware operates:**
A) In memory without writing files to disk
B) Only on USB devices
C) By infecting documents
D) Through email attachments
**Answer:** A
**Explanation:** Fileless malware avoids detection by staying in memory.

9.

**Which malware family includes scripts like PowerShell or JavaScript used maliciously?**
A) Script-based malware
B) Virus
C) Worm
D) Ransomware
**Answer:** A
**Explanation:** Malicious scripts perform attacks without executables.

10.

**Botnets are:**
A) Networks of compromised computers controlled remotely
B) Types of ransomware
C) Antivirus tools
D) Adware software
**Answer:** A
**Explanation:** Botnets coordinate attacks using many infected machines.

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# Medium (20 Questions)

11.

**What is the main goal of polymorphic malware?**
A) To change its code to evade signature detection
B) To encrypt files
C) To display ads
D) To record keystrokes
**Answer:** A
**Explanation:** Polymorphic malware modifies its appearance.

12.

**Malware-as-a-Service refers to:**
A) Renting malware tools on dark web marketplaces
B) A free antivirus program
C) Legitimate software updates
D) Network firewall rules
**Answer:** A
**Explanation:** Malware developers sell services for attacks.

13.

**Which malware trend involves unauthorized mining of cryptocurrencies?**
A) Cryptojacking
B) Ransomware
C) Worms
D) Spyware
**Answer:** A
**Explanation:** Cryptojacking uses victim devices to mine coins.

14.

**What is a logic bomb?**
A) Malware triggered by specific conditions or dates
B) Virus that spreads via email
C) Spyware monitoring keystrokes
D) Advertising software
**Answer:** A
**Explanation:** Logic bombs activate under predefined triggers.

15.

**Which mobile OS is most targeted by malware?**
A) Android
B) iOS
C) Windows Mobile
D) Blackberry OS
**Answer:** A
**Explanation:** Android's open ecosystem attracts more malware.

16.

**In static malware analysis, what is examined?**
A) Code and file structure without running the malware
B) Network traffic during execution
C) User activity logs
D) System registry changes during attack
**Answer:** A
**Explanation:** Static analysis studies malware code safely.

17.

**Dynamic malware analysis involves:**
A) Running malware in a sandbox to observe behavior
B) Disassembling code
C) Scanning files for signatures
D) Encrypting user files
**Answer:** A
**Explanation:** Dynamic analysis monitors real-time effects.

18.

**Which tool is commonly used for malware sandboxing?**
A) Cuckoo Sandbox
B) Wireshark
C) Nmap
D) Metasploit
**Answer:** A
**Explanation:** Cuckoo analyzes malware safely in isolated environments.

19.

**Supply chain attacks involve:**
A) Infecting software updates or third-party providers
B) Email phishing attacks
C) Password cracking
D) Firewall misconfiguration
**Answer:** A
**Explanation:** Supply chain attacks target trusted software sources.

20.

**Which malware family uses macros in office documents to infect systems?**
A) Macro malware
B) Worm
C) Rootkit
D) Botnet
**Answer:** A
**Explanation:** Malicious macros execute harmful code in documents.

21.

**AI-powered malware can:**
A) Adapt its behavior to avoid detection

B) Encrypt data faster
C) Scan networks for open ports
D) Update antivirus definitions
**Answer:** A
**Explanation:** AI malware learns to bypass defenses.

22.

**Which malware is designed to provide remote control of an infected system?**
A) RAT (Remote Access Trojan)
B) Worm
C) Virus
D) Keylogger
**Answer:** A
**Explanation:** RATs allow attackers full control remotely.

23.

**Which of these is NOT a characteristic of ransomware?**
A) Encrypts user data
B) Demands payment for decryption
C) Spreads via self-replication like a worm
D) Often delivered through phishing emails
**Answer:** C
**Explanation:** Ransomware does not self-replicate like worms.

24.

**Adware is primarily:**
A) Software that displays unwanted advertisements
B) Malware that steals passwords
C) A type of worm
D) A virus that infects executables
**Answer:** A
**Explanation:** Adware bombards users with ads.

25.

**Which of these malware analysis techniques involves monitoring network traffic?**
A) Dynamic analysis
B) Static analysis
C) Signature scanning
D) Encryption
**Answer:** A
**Explanation:** Dynamic analysis observes network behavior.

26.

**Which of the following is a common indicator of malware infection?**
A) Unexpected system slowdowns
B) Faster internet speeds
C) Increased battery life
D) Improved application performance

**Answer:** A
**Explanation:** Malware often consumes resources, slowing systems.

27.

**What is a Trojan downloader?**
A) Malware that downloads other malicious programs
B) Antivirus software
C) Network scanning tool
D) Firewall rule
**Answer:** A
**Explanation:** It downloads additional malware after infection.

28.

**Fileless malware commonly abuses:**
A) Legitimate system tools like PowerShell
B) USB devices only
C) Email attachments
D) Document macros exclusively
**Answer:** A
**Explanation:** Uses trusted tools to evade detection.

29.

**Which type of malware can hide itself deep in the OS and avoid detection?**
A) Rootkit
B) Virus
C) Worm
D) Adware
**Answer:** A
**Explanation:** Rootkits modify OS to stay hidden.

30.

**What is the role of sandboxing in malware analysis?**
A) To safely execute and observe malware behavior
B) To delete infected files immediately
C) To encrypt malware code
D) To prevent malware from running
**Answer:** A
**Explanation:** Sandboxes isolate malware from real systems.

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## 🔴 Hard (10 Questions)

31.

**Polymorphic malware uses what technique to evade detection?**
A) Changing its code on every infection

B) Encrypting user files
C) Installing keyloggers
D) Hijacking sessions
**Answer:** A
**Explanation:** It morphs code to avoid signature detection.

32.

**Which malware analysis method can be dangerous because the malware is executed?**
A) Dynamic analysis
B) Static analysis
C) Signature scanning
D) Heuristic analysis
**Answer:** A
**Explanation:** Executing malware poses risks without isolation.

33.

**Which malware is known for exploiting zero-day vulnerabilities?**
A) Advanced Persistent Threats (APTs)
B) Adware
C) Worms
D) Keyloggers
**Answer:** A
**Explanation:** APTs use unknown flaws for targeted attacks.

34.

**In malware analysis, "YARA rules" are used for:**
A) Detecting and classifying malware based on patterns
B) Encrypting files
C) Monitoring network packets
D) Generating passwords
**Answer:** A
**Explanation:** YARA helps identify malware signatures.

35.

**What is the main difference between viruses and worms?**
A) Viruses require host files; worms self-replicate independently
B) Worms require host files; viruses self-replicate
C) Viruses encrypt files; worms do not
D) Worms only work on mobile devices
**Answer:** A
**Explanation:** Viruses infect files; worms spread on networks.

36.

**Which malware evasion technique tries to detect if it's running in a virtual machine?**
A) Anti-VM techniques
B) Polymorphism
C) Rootkit hiding
D) Keylogging

**Answer:** A
**Explanation:** Malware avoids sandbox analysis by detecting VMs.

37.

**Which analysis tool helps in disassembling malware binaries?**
A) IDA Pro
B) Wireshark
C) Metasploit
D) Nmap
**Answer:** A
**Explanation:** IDA Pro is a popular disassembler.

38.

**Which malware analysis phase involves unpacking or decrypting the malware code?**
A) Unpacking phase
B) Execution phase
C) Monitoring phase
D) Signature scanning
**Answer:** A
**Explanation:** Malware is often packed/encrypted to evade analysis.

39.

**What type of malware attack targets supply chains?**
A) Infecting third-party software updates
B) Keylogging
C) Phishing
D) Ransomware
**Answer:** A
**Explanation:** Supply chain attacks infect trusted software sources.

40.

**Which of these is NOT a typical goal of malware?**
A) Improve system performance
B) Steal data
C) Damage or disrupt systems
D) Generate revenue for attackers
**Answer:** A
**Explanation:** Malware does not improve system performance.